STAT 4C03 Question 1

a)

```
happy = read.csv("Happiness.csv")
head(happy)
```

```
##   happy money sex love work
## 1    10    36   0    3    4
## 2     8    47   1    3    1
## 3     8    53   0    3    5
## 4     8    35   1    3    3
## 5     4    88   1    1    2
## 6     9   175   1    3    4
```

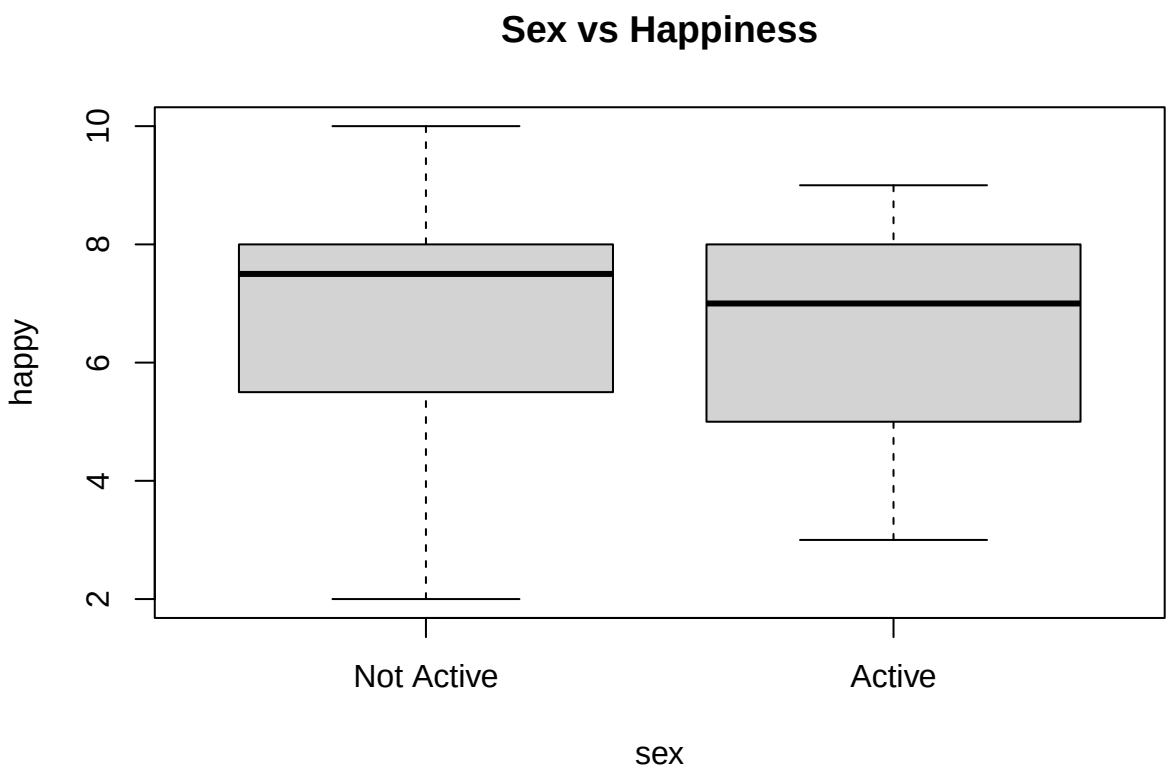
```
# Convert covariates to factors
happy$sex <- factor(happy$sex, levels = c(0, 1), labels = c("Not Active", "Active"))
happy$love <- factor(happy$love, labels = c("Lonely", "Some", "Deep"))
happy$work <- factor(happy$work)
head(happy)
```

```
##   happy money        sex   love work
## 1    10    36 Not Active   Deep    4
## 2     8    47     Active   Deep    1
## 3     8    53 Not Active   Deep    5
## 4     8    35     Active   Deep    3
## 5     4    88     Active Lonely    2
## 6     9   175     Active   Deep    4
```

```
# Plots
plot(happy~money, data = happy, main = "Money vs Happiness")
```




```r
boxplot(happy ~ sex,  data = happy, main = "Sex vs Happiness")
```




```r
boxplot(happy ~ love, data = happy, main = "Love vs Happiness")
```




```
boxplot(happy ~ work, data = happy, main = "Work vs Happiness")
```



The scatterplot of money against happy shows no discernible trend, the points are essentially scattered horizontally across the happiness scale. Sex shows not active individuals tending toward higher happiness scores. money looks like the weakest predictor, with no apparent relationship to happiness.

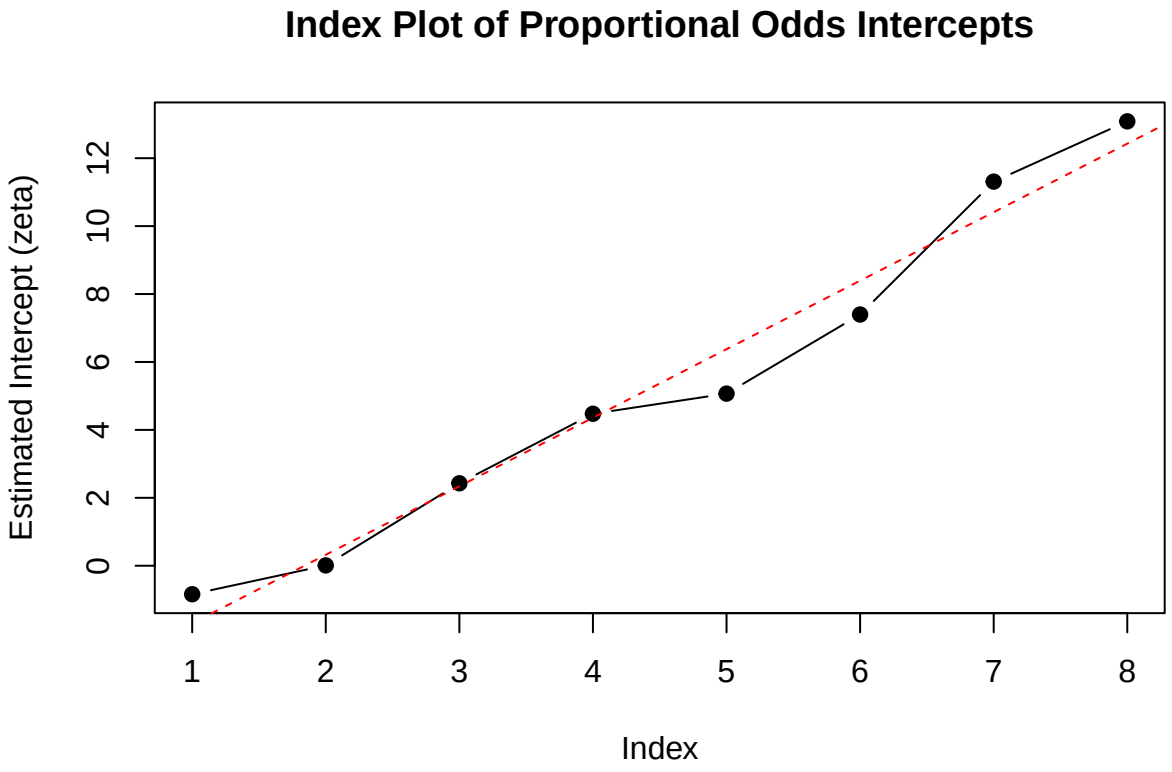
b)

```r
# happy to ordered factor and modeling
library(MASS)
happy$happy <- as.ordered(happy$happy)
m1 <- polr(happy ~ money + sex + love + work, data = happy, Hess=TRUE)
summary(m1)
```

```
## Call:
## polr(formula = happy ~ money + sex + love + work, data = happy,
##     Hess = TRUE)
##
## Coefficients:
##              Value Std. Error   t value
## money      0.01783    0.01087  1.640201
## sexActive -1.02518    0.93630 -1.094930
## loveSome   3.45767    1.56125  2.214685
## loveDeep   7.85065    1.85206  4.238877
## work2     -1.18911    1.68767 -0.704588
## work3      0.01569    1.58057  0.009929
## work4      1.84631    1.53697  1.201261
## work5      0.64777    2.14985  0.301308
##
## Intercepts:
##      Value   Std. Error t value
## 2|3  -0.8392  1.8387    -0.4564
## 3|4   0.0099  1.7713     0.0056
## 4|5   2.4279  2.0150     1.2049
## 5|6   4.4745  2.1064     2.1243
## 6|7   5.0676  2.1243     2.3855
## 7|8   7.3974  2.2303     3.3168
## 8|9  11.3106  2.5925     4.3628
## 9|10 13.0851  2.7917     4.6872
##
## Residual Deviance: 90.47841
## AIC: 122.4784
```

The proportional odds model with k ordered response categories has k−1 threshold parameters. The happy variable takes integer values but, in this sample of 39 students, one value on the 1–10 scale does not appear, so only 9 distinct levels are observed, yielding $9-1 = 8$ estimated intercepts.

```r
# Index plot of estimated intercepts (zeta)
plot(m1$zeta, type = "b", pch = 19,
     xlab = "Index", ylab = "Estimated Intercept (zeta)",
     main = "Index Plot of Proportional Odds Intercepts")
abline(lm(m1$zeta ~ seq_along(m1$zeta)), col = "red", lty = 2)
```



The intercepts are roughly linear at lower indices but deviate noticeably in the middle and upper range: the small gap between indices 4–5 suggests those happiness levels are nearly indistinguishable, while the large jump at 5–6 indicates a more pronounced threshold. This implies the latent happiness scale is not uniformly spaced.

c)

```
# Individual 1: lonely, not sexually active, no job, income 30
nd1 <- data.frame(
  money = 30,
  sex   = factor("Not Active", levels = levels(happy$sex)),
  love  = factor("Lonely",     levels = levels(happy$love)),
  work  = factor(1,            levels = levels(happy$work))
)

# Individual 2: same but deep feeling of belonging
nd2        <- nd1
nd2$love   <- factor("Deep", levels = levels(happy$love))

predict(m1, nd1, type = "probs")
```

```
##            2            3            4            5            6            7
## 2.019666e-01 1.697279e-01 4.974154e-01 1.118077e-01 8.445792e-03 9.591555e-03
##            8            9           10
## 1.024218e-03 1.735533e-05 3.543881e-06
```

```r
predict(m1, nd2, type = "probs")
```

```
##            2            3            4            5            6            7
## 9.856451e-05 1.318024e-04 2.349225e-03 1.704920e-02 1.533345e-02 2.363308e-01
##            8            9           10
## 6.777815e-01 4.190881e-02 9.016585e-03
```

Individual 1 (lonely): the distribution is concentrated at the lower end, with most probability mass at category 4 (~50%), followed by categories 2 and 3. Virtually no probability is assigned above category 6. This individual is predicted to have low happiness.

Individual 2 (deep belonging): the distribution shifts dramatically to the upper end, category 8 alone captures ~68% of the probability, with category 7 taking another ~24%. There is essentially zero probability below category 6.

d)

```r
# test all covariates
drop1(m1, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## happy ~ money + sex + love + work
##        Df    AIC     LRT  Pr(>Chi)
## <none>    122.48
## money   1 123.31  2.8309   0.09247 .
## sex     1 121.68  1.2056   0.27220
## love    2 149.91 31.4332 1.494e-07 ***
## work    4 123.81  9.3336   0.05328 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We drop money which has the largest none-significative p-value.

```r
m3 <- polr(happy ~ love + work + sex, data = happy)
drop1(m3, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## happy ~ love + work + sex
##        Df    AIC    LRT  Pr(>Chi)
## <none>    123.31
## love    2 151.92 32.609 8.299e-08 ***
## work    4 126.72 11.406   0.02236 *
## sex     1 122.22  0.909   0.34032
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We drop sex which has the largest none-significative p-value.

```
m4 <- polr(happy ~ love + work , data = happy)
drop1(m4, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## happy ~ love + work
##        Df    AIC    LRT  Pr(>Chi)
## <none>    122.22
## love    2 151.41 33.190 6.208e-08 ***
## work    4 126.01 11.792   0.01897 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m4, m1)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: happy
##                       Model Resid. df Resid. Dev   Test    Df LR stat.
## 1               love + work        25   94.21851
## 2 money + sex + love + work        23   90.47841 1 vs 2     2 3.740104
##     Pr(Chi)
## 1
## 2 0.1541156
```

Conclusion:

The p-value $= 0.15 > 0.05$, we have no evidence at 5% level that the reduced model (m4) is worser than the full model. The dropped covariates doesn't contribute significantly to the model.

e)

```
happy_n       <- happy
happy_n$love  <- as.numeric(happy_n$love)
happy_n$work  <- as.numeric(happy_n$work)

m4_num <- polr(happy ~ love + work, data = happy_n)
anova(m4_num, m4)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: happy
##         Model Resid. df Resid. Dev   Test    Df LR stat.   Pr(Chi)
## 1 love + work        29   99.62877
## 2 love + work        25   94.21851 1 vs 2     4 5.410258 0.2477312
```

Conclusion:

We failed to reject the null hypothesis with a p-value of 0.24. The simpler numeric model is adequate and more parsimonious. Model selection. Subject to the LRT result, we would prefer m4_num (love and work as numeric scores): it has the same explanatory power with 4 fewer parameters.

f)

```
nd1_n <- data.frame(love = 1, work = 1)
nd2_n <- data.frame(love = 3, work = 1)

predict(m4_num, nd1_n, type = "probs")
```

```
##            2            3            4            5            6            7
## 4.399790e-01 2.489206e-01 2.879241e-01 1.877600e-02 1.733990e-03 2.284384e-03
##            8            9           10
## 3.658066e-04 1.266264e-05 3.392172e-06
```

```
predict(m4_num, nd2_n, type = "probs")
```

```
##            2            3            4            5            6            7
## 0.0007785174 0.0014126808 0.0379295327 0.1431405903 0.0873192064 0.4513338375
##            8            9           10
## 0.2621540612 0.0125226486 0.0034089252
```

Removing money and sex from the model shifts probability mass further toward the extremes, the final model is more decisive.